$$(\because 0 = \sum_{j=1}^{k} 0 \cdot \sigma^{k-j} \text{ and Lemma 7.7.12})$$

$\Rightarrow h^{-1}(s_j) = h^{-1}(t_j),\ 1 \leq j \leq k$

$\Rightarrow s_j = t_j,\ 1 \leq j \leq k \qquad (h^{-1}\ is\ one\text{–}to\text{–}one)$

$\Rightarrow (s_1, s_2, \cdots, s_k) = (t_1, t_2, \cdots, t_k) \qquad$ (Lemma 7.3.4)

$\Rightarrow (s_1, s_2, \cdots, s_k) = (t_1, t_2, \cdots, t_l) \qquad (\because l = k)$

$\Rightarrow w = w'$

Hence, f is one-to-one.

Therefore, $\Sigma^+ \sim \mathbf{N}$ ■

Theorem 7.7.14: There exists a one-variable number-theoretic function which is not computed by any program in $\mathcal{P}_\Sigma$.

Proof: (proof by contradiction) Suppose to the contrary that every function in $\psi^{(1)}$ is computed by a program in $\mathcal{P}_\Sigma$.

By Lemma 7.7.13, there exists a function $f : \Sigma^+ \overset{1-1}{\underset{onto}{\rightarrow}} \mathbf{N}$.

Let $g = f^{-1}$. Then $g : \mathbf{N} \overset{1-1}{\underset{onto}{\rightarrow}} \Sigma^+$. (Theorem 6.3.2)

For each $i \in \mathbf{N}$, let $F_i : \mathbf{N} \rightarrow \mathbf{N}$ such that

$$F_i(x) = \begin{cases} the\ positive\ integer\ n & if\ \begin{array}{l} g(i)\ is\ a\ valid\ program,\ and\ g(i)\ outputs \\ a\ positive\ integer\ n\ on\ input\ x \end{array} \\ 1 & otherwise \end{cases}$$

Therefore, the string $g(i)$ computes the function F_i if $g(i)$ is a valid program which halts on every positive integer input and produces a positive integer output. When $g(i)$ is not a valid program, or if it is but it enters an infinite loop or produces an output that is not a positive integer on some input x, then we assume that it produces the output 1 on that input x. In this way, we generously regard every string in Σ^+ as a valid program. We shall show that, even with such generosity, there is at least one function in $\psi^{(1)}$ which is not computed by any string in Σ^+ and hence by any program in $\mathcal{P}_\Sigma$.

Now, define a function $d : \mathbf{N} \rightarrow \mathbf{N}$ such that:

$$d(x) = F_x(x) + 1, \forall x \in \mathbf{N}$$

Figure 2 depicts the construction of function d based on the family of functions $\{F_i\}_{i \in \mathbf{N}}$. The technique underlying the proof is again the diagonalization technique.

Since d is a function from $\mathbf{N}$ to $\mathbf{N}$, $d \in \psi^{(1)}$. Moreover, as we have assumed that every function in $\psi^{(1)}$ is computed by a program in $\mathcal{P}_\Sigma$ (hence a string in Σ^+) and $\{F_i\}_{i \in \mathbf{N}}$ consists of all the functions that are computed by the strings in Σ^+, there must exist an i such that $d = F_i$.

But then,

$d = F_i$

$\Rightarrow d(x) = F_i(x), \forall x \in \mathbf{N} \qquad$ (Lemma 6.1.10)

$\Rightarrow d(i) = F_i(i) \qquad$ (UI)

$\Rightarrow F_i(i) + 1 = F_i(i) \qquad (d(i) = F_i(i) + 1)$

$\Rightarrow 1 = 0,$ a contradiction!

Hence, there exists a one-variable number-theoretic function which is not computed by any program in $\mathcal{P}_\Sigma$. ■

Figure 2

	1	2	3	...	i	...
F_1	$F_1(1)$	...	...	...	...	...
F_2	...	$F_2(2)$	...	...	...	...
F_3	...	...	$F_3(3)$	...	...	...
$\vdots$	...	...	...	$\ddots$	...	...
F_i	...	...	...	...	$F_i(i)$	...
$\vdots$	...	...	...	...	...	$\ddots$
d	$F_1(1)+1$	$F_2(2)+1$	$F_3(3)+1$	...	$F_i(i)+1$	...

7.8 Exercises

1. Prove that if R is a symmetric relation, then so is $R^k, \forall k > 0$.
2. Let A be a nonempty set, and 2^A be the set of all functions from A to $\{0,1\}$. Prove that $\mathcal{P}(A) \sim 2^A$, where $\mathcal{P}(A)$ is the power set of A.
3. Let A and B be any two sets. Prove that $A \times B \sim B \times A$.
4. Let A and B be any two sets. Prove that $A \times (B \cup C) \sim (A \times B) \cup (A \times C)$.
5. Prove that if $(X - Y) \sim (Y - X)$, then $X \sim Y$.
6. Prove that if $A \sim C$ and $B \sim D$, then $A \times B \sim C \times D$.
7. Prove that if $A \sim B$, then $A^C \sim B^C$ for any set C.
8. Let A, B, C be sets such that $B \cap C = \emptyset$. Prove that $A^{B \cup C} \sim A^B \times A^C$.
9. Let A, B, C be sets. Prove that $(A \times B)^C \sim A^C \times B^C$.
10. Prove that the open interval $(0,1) = \{x \in \mathbf{R} \mid 0 < x < 1\}$ is an infinite set.
11. Let $\mathfrak{B} = \{\{b_i\}_{i=1}^{\infty} \mid b_i \in \{0,1\}\}$. i.e. $\mathfrak{B}$ is the set of all infinite sequences of 0's and 1's. Prove that $\mathfrak{B}$ is uncountable.
12. Determine $F_i(x)$ and $d(i)$ for each of the following $g(i)$ where g is defined in Lemma 7.7.14.
    (a) $g(i) =$ "**program** h(input,output); **var** x:**integer**; **begin** read(x); write(2*x) **end**."
    (b) $g(i) =$ "**program** h(input,output); **var** x:**integer**; **begin** read(x); **repeat** x:=x **until** (false) **end**."
    (c) $g(i) =$ "**program** h(input,output); **var** x:**integer**; **begin** anjgishgragjskklsjjbumijk **end**."
    (d) $g(i) =$ "**program** h(input,output); **var** x:**integer**; **begin** read(x); **if** (x<50) **then** write(x-30) **elseif** (x<1000) **then** write(x/2) **else** write("too large") **end**."
13. Use the diagonalization method to prove that there exists a one-variable number theoretic function which cannot be defined mathematically using a finite set of symbols.
14. Let A and B be two finites sets. Prove that $|A| \leq |B|$ if and only if $\exists f : A \overset{1-1}{\rightarrow} B$.